

Suresh P

Computer Science graduate with internship experience in AI/ML, computer vision, and deep learning. Built real time machine learning applications including Indian Sign Language Recognition and Automatic Number Plate Recognition using TensorFlow, OpenCV, MediaPipe, and YOLO. Passionate about developing practical AI solutions for real-world problems.

Experience

AI / ML Intern

Jun 2026 – Aug 2026

HCL Technologies

Deep Learning Video Processing / Animation System

- Implemented data preprocessing, feature engineering, model training and validation pipelines using Python and Scikit-learn.
- Collaborating with senior engineers on model training, validation, and optimization workflows using industry-standard ML frameworks.

Key Technologies Used: Python, Scikit-learn, PyTorch, Matplotlib, ImageIO

Python Intern

June 2025

Entudio Pvt. Ltd

Backend Development & API Optimization

- Refactored Python data-processing modules, reducing execution time by 30% through algorithm and memory optimizations.
- Designed and developed REST APIs using Flask for automated JSON based data exchange, reducing manual processing effort by 40%.

Key Technologies Used: Python, Flask, REST API, JSON, NumPy

Mobile App Dev Intern

March 2025

Hicup Innovations

Mobile Application Development (Flutter)

- Developed Flutter frontend interfaces and integrated Firebase for real-time database management.

Key Technologies Used: Flutter, Dart, Firebase

Achievements

- Earned Elite Certification (75%) in **NPTEL Cloud Computing** (IIT Kharagpur).
- Published a research paper on Indian Sign Language Recognition and Translation System in the **“Journal of Ubiquitous Computing and Communication Technologies”**.

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 [Suresh P](#)

 [GitHub](#)

Technical Skills

- Python
 - HTML
 - CSS
 - JavaScript
 - Machine Learning
 - Deep Learning
 - Sci-kit learn
 - Random Forest
 - CNN
 - Numpy
 - Pandas
 - Data Visualization
 - OpenCV
 - Tensorflow
 - YOLO
 - BiLSTM
 - EasyOCR
 - Computer Vision
 - Matplotlib
 - Flask
 - REST APIs

Language

English

Professional Working

Tamil

Native Language

Projects

Indian Sign Language Recognition & Translation

[GitHub](#)

- Engineered a hardware-free ISL recognition system achieving 95.8% static sign accuracy using a hybrid inference engine — Random Forest as primary classifier with CNN fallback, and a 3-layer BiLSTM for dynamic gesture sequences.
- Engineered a MediaPipe-based feature extraction pipeline generating 126 3D landmark features per frame for robust static and dynamic gesture recognition.
- Integrated a locally deployed Phi-3 language model through LM Studio for grammar-aware sentence generation and multilingual speech synthesis across 11 languages.

Tools: [Computer Vision](#), [Deep Learning](#), [MediaPipe](#), [TensorFlow](#), [Flask](#).

Automatic Number Plate Recognition (ANPR)

[GitHub](#)

- Developed an Automatic Number Plate Recognition system that detects vehicle license plates using YOLOv5 and extracts registration numbers with EasyOCR.
- Built an end-to-end computer vision pipeline integrating image preprocessing, plate localization, OCR, and real-time visualization using OpenCV.
- Implemented a scalable inference pipeline supporting both still-image and video-based vehicle detection.

Tools: [YOLO](#), [EasyOCR](#), [OpenCV](#), [Python](#).

Gourmet AI – Smart Recipe Generation System

[GitHub](#)

- Developed an AI-powered recipe generation system that suggests dishes based on user-provided ingredients using generative AI APIs. Built a backend service for real-time request handling and structured recipe outputs.
- Implemented retry mechanisms to handle API rate-limit errors (HTTP 429) and improved system reliability through efficient error management. Designed an interactive frontend for instant recipe suggestions, with planned integration of camera-based ingredient detection.

Tools: [Generative AI](#), [FastAPI](#), [JavaScript](#), [REST APIs](#).

PPE Guard – Real-Time Safety Compliance System

[GitHub](#)

- Developed an AI-powered Personal Protective Equipment (PPE) detection system for automated workplace safety audits, capable of identifying six safety gear classes using a custom-trained YOLOv8 Nano model.
- Engineered a low-latency live streaming architecture using FastAPI and WebSockets, implementing frame-draining and thread-pooling optimizations to ensure real-time webcam inference.
- Built a modular Cyber-Industrial UI using JavaScript and Tailwind CSS, supporting dynamic compliance policies across live streams, static images, and video file uploads.

Tools: [YOLOv8](#), [FastAPI](#), [OpenCV](#), [WebSockets](#), [Python](#), [JavaScript](#).

Education

B.E. Computer Science & Engineering

2022 – 2026

Grade: 8.1/10

VV College of Engineering

HSC

2021 – 2022

Grade: 72.5%

E.R.S. Higher Secondary School

SSLC

2019 – 2020

Grade: 76.8%

E.R.S. Higher Secondary School